Liquidity, Activism Disclosure, and Takeover Premia

A Theory of Exit, Voice, and Corporate Control

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This draft: 30 August 2026

Abstract

A large shareholder who accumulates quietly must eventually reveal herself. Cross five percent of a voting class and a filing comes due, in a window February 2024 cut from ten calendar days to five business days. This paper treats that rule as the market's partition. The threshold decides who files; the window, how much accumulation stays hidden first. In a two-round model of a privately informed blockholder facing bidder entry, tightening the threshold margin attenuates the sensitivity of takeover premia to liquidity at fixed policies, while tightening the window margin attenuates if and only if a condition the paper states holds. The two margins are not the same instrument. Filers moved: in a seeded filing sample, median filing delay fell from seven to five business days and the share filed inside the new deadline rose from 36 to 76 percent. A pre-specified design takes the partition to the data.

1 Introduction

A blockholder who learns bad news about a company can sell, sit still, or engage. If she engages and buys, the law eventually makes her say so. In the United States the saying-so is governed by Schedule 13D of the Williams Act: cross 5 percent of the voting class and you must file within a window that, from February 5, 2024, runs five business days instead of the ten calendar days that had stood since 1968 (U.S. Securities and Exchange Commission, 2023). The rule has two separable parts, and this paper treats each as its own policy coordinate. The *threshold margin* decides which stakes trigger a filing at all; the United States sets it at 5 percent, the United Kingdom at 3 percent with rungs at every point above. The *window margin* decides how much accumulation can happen before the filing lands; the February 2024 acceleration shortened it, from ten calendar days to five business days. Neither margin is scenery. Both move what the market sees while a large trader is still trading, and what the market sees moves what a bidder is willing to pay.

The paper's identity is that the disclosure rule *is* the market's partition. Given a stake threshold τ and a filing window T, the histories that reach the control node split into two cells. In the *flagged* cell the filing has landed and reveals the stake and the intent to engage. In the *pooled* cell it has not, and the market must infer the blockholder's presence from order flow alone. Nothing else in the model performs that split. The question the paper asks is how noise-trading intensity (liquidity, the model's κ) moves bidder entry and the expected takeover premium once the rule has drawn that line, and how the answer changes when the line is redrawn through either margin.

The answer differs across the margins, and saying so precisely is the paper's main theoretical contribution. Earlier work on liquidity and blockholder governance gives one-parameter answers: liquidity funds monitoring (Maug, 1998), liquidity is irrelevant under free entry (Kahn and Winton, 1998), liquidity blunts the disciplining threat of exit by making exit less informative (Admati and Pfleiderer, 2009). The disclosure rule sits outside all of them. Maug notes that pre-trade disclosure would destroy camouflage and moves on. The trading-and-takeover literature has the pieces without the rule: Kyle and Vila (1991) split the market's inference between pooled and revealed trading but disclaim disclosure requirements by assumption; Back et al. (2018) have a fixed horizon at which the stake becomes known and show that shortening it is isomorphic to reducing noise-trading volatility, so for them the window is a rescaling, not a mechanism; Cetemen et al. (2026) place a whole multi-trader game inside an unmodelled pre-disclosure window. No paper in the competitor set combines the two margins of the rule, liquidity, and a control outcome. The cell this paper occupies is the partition itself: an on-path flagged/pooled split keyed to the rule's parameters, a stake trigger and a legal deadline, feeding an endogenous bidder-entry decision.

Three results carry the paper, each stated with the hypotheses it uses.

Existence (Proposition 1). A cutoff perfect Bayesian equilibrium over complete contingent plans exists at every $\kappa \in [0, 1]$: the blockholder picks a plan from a finite ordered menu (a whole stake path plus an engagement flag, not a single order), and equilibrium play is a weakly ordered cutoff vector in the signal. The statement is conditional on named hypotheses about the equilibrium object, and the paper records the conditionality rather than burying it: two of those hypotheses are measured to fail at the calibration the accompanying implementation runs, so at that calibration the proposition asserts nothing. Nonexistence is neither claimed nor shown; twenty-three of twenty-seven equilibrium-sweep nodes converge.

Attenuation (Theorem 1). At fixed policies, the liquidity sensitivity of the expected engagement-related premium factors exactly as $\mathcal{S} = (1-\Omega)\mathcal{S}_P$: the pooled cell's sensitivity, scaled by the pooled cell's weight, where Ω is the probability the flag lands. Tightening the threshold margin moves mass from the pooled cell into the flagged one and attenuates, with no dominance condition. Tightening the window margin attenuates if and only if $W_T C_T \leq 1$: the weight leg $W_T \leq 1$ is proved, but the composition leg C_T is unsigned, because a shorter window also changes *who* is left in the pool. The window margin is therefore an "if and only if", not a sign, which is exactly why the February 2024 acceleration is an empirical question rather than a corollary.

From fixed policies to equilibrium (Proposition 2). Once the cutoff vector moves with liquidity, the term the factorisation discards returns. The fixed-policy attenuation sign survives in equilibrium on a dominance-and-contraction region carried as a named hypothesis; the region is not exhibited, and the grid record is that eighteen of eighty nodes satisfy the two pointwise inequalities, in a largest contiguous block of nine.

The numerical layer reads these results against the implemented calibration, and the paper reports what it says without smoothing. Cutting the window from $T = 10$ to $T = 5$ cuts the premium's liquidity sensitivity to between 18 and 77 percent of its long-window value at every checked node, and the composition leg carries almost all of it, so at that calibration the model's directional prediction for the 2024 acceleration is attenuation. But the chord restriction behind the threshold-margin leg and the composition ratios is measured to fail at the same calibration: the pooled posterior law carries between 23 and 767 distinct posterior values where the restriction's representation asks for three, at every one of 180 non-degenerate nodes. The conditional results remain proved as conditionals; at that calibration they say nothing about the implemented pooled cell, and the specification for the empirics carries both possible signs rather than assuming the theory's direction. Section 5.6 carries the full summary, including the corner caveat on the long window, the interior re-run that points the same way, and the route that re-run travels.

The empirical half of the paper is built on the February 2024 acceleration and is honest about what has been run. Executed descriptives from a seeded sample of initial 13D filings show the median trigger-to-filing delay falling from 7.0 business days before the rule to 5.0 after it, and the share of filings inside the new five-business-day deadline rising from 35.7 to 75.6 percent (Section 6). That is the first stage, and it lines up with what the Commission itself projected. The rest of the empirics is a written specification, fixed before estimation, with six substantive legs: a *timing split* that tests the partition directly (the run-up from the 5 percent crossing to the filing should move with liquidity, the filing-day jump should not, and the two slopes differ); a *bindingness dose* that splits filers by how hard the new deadline binds them; a *stake at filing*, the model's own object, read against the shorter window; a *bounded null*, arithmetic on the Commission's own tables that caps the accumulation channel's effect on the twelve-month bid hazard at about three percentage points; a *matched difference-in-differences* on that hazard, which cannot separately detect even the ceiling; and a *bidder-entry-by-liquidity* leg, which adds the liquidity interaction to the control-outcome test. The timing split is the specification's default headline; final selection remains pending in Section 6.3.

The paper proceeds as follows. Section 2 sets out the institutional detail and positions the paper against the literature, and Section 3 builds the two-round model and states the standing hypotheses. Existence is the subject of Section 4; liquidity and control outcomes follow in Section 5, from the partition lemmas through the attenuation theorem, the general-equilibrium proposition, and the numerical records. Section 6 presents the motivating facts and the written specification for the full empirics, and Section 7 concludes. The proofs sit in the online appendix.

2 Institutional setting and related literature

2.1 The disclosure rule and its two margins

Section 13(d) of the Williams Act of 1968 requires any person or group acquiring beneficial ownership of more than 5 percent of a class of registered voting equity to file a Schedule 13D disclosing the stake and its purpose. From 1968 until 2024 the initial deadline was ten calendar days. The SEC proposed shortening it in February 2022, adopted the amendments in October 2023, and made them effective February 5, 2024: the initial 13D deadline became five business days, amendments on material change became two business days, and the parallel changes for Schedule 13G took effect September 30, 2024 (U.S. Securities and Exchange Commission, 2022; U.S. Securities and Exchange Commission, 2023). Item 4 of the form asks the filer to state its purpose; control-intent filers use the 13D, while the shorter 13G is open to three eligible classes (exempt investors, qualified institutional investors, and passive holders below 20 percent who certify no control intent). Which of the two a given block lands in is a choice the rule frames rather than an accident of the holder, and eligibility for the short form is itself a type variable rather than a random one.

Both margins of the rule vary across jurisdictions, and the variation is not subtle. The United Kingdom sets the initial threshold at 3 percent with notification at every additional percentage point, within two trading days. The EU Transparency Directive sets a ladder at 5 percent and above with four trading days, and several member states start at 3 percent. The threshold margin is therefore a cross-section object and the window margin a time-series one. Other jurisdictions have moved their windows too, the EU cutting its Transparency Directive deadline to four trading days in 2013 among them; what recommends the February 2024 acceleration as an anchor is that it is a dated change to one margin, in one jurisdiction, with a clean before and after. That is why it anchors this paper and why it has begun to anchor others (Polk et al., 2024; Trivedi, 2026).

Two institutional facts discipline the model. First, the window is a *legal deadline attached to a partition*, not a random horizon: the clock starts when the stake crosses the threshold, the filing lands at most T days later, and between the crossing and the filing the market prices the stock without the flag. The model takes the deadline itself as the filing date, so filing early is outside what it claims. Second, the pooled cell is pooled *for the price-setting market.* Zeng (2026) documents that a target's own managers learn of the accumulation through stock-surveillance vendors before the filing, and that the price run-up begins on the trigger date. The cell is leaky at its edges, which is a measurement problem for the empirics, not a failure of the partition for the market makers who set prices. On the delay distribution itself, Bebchuk et al. (2013) show filers bunching at the old deadline: 45.3 percent of filings by engaging investors in their sample land in the eight-to-ten-*business*-day bucket and roughly a fifth on the tenth business day itself, their delays being counted in business days against a deadline the rule states in calendar days. So the window binds for a large share of filers, which is what makes the February 2024 acceleration a treatment rather than a formality.

2.2 Exit, voice, and liquidity

The classic trade-off originates with Hirschman (1970). Coffee (1991) and Bhide (1993) argue that liquidity weakens governance by letting blockholders sell instead of fight; Maug (1998) reverses the logic: liquid markets let a monitor recoup the costs of engagement through informed trading, and with an endogenous stake liquidity raises equilibrium monitoring. Kahn and Winton (1998) show the sign of the trading motive depends on whether engagement deepens the blockholder's information advantage; Faure-Grimaud and Gromb (2004) make price informativeness and insider incentives complements; Admati and Pfleiderer (2009) make exit itself a disciplining threat. On the empirical side, Edmans, Fang, et al. (2013) find more liquid targets shift from voice to exit, and Norli et al. (2015) find liquidity predicts the incidence of costly voice.

Two limitations of this strand shape the present paper. The four classics use four different liquidity parameters (camouflage, noise volume, informativeness, exit noise) with three different signs, so "liquidity and governance" is not one question. And none of the four has a bidder, an offer, or a premium: takeovers appear in Maug (1998) only as a relabelled intervention technology, Kahn and Winton (1998) and Faure-Grimaud and Gromb (2004) set control transfers aside in terms, and Admati and Pfleiderer (2009) simply has no bidder. The object this paper studies, the sensitivity of an expected takeover premium to liquidity and what the disclosure rule does to that sensitivity, is not stated in any of them.

2.3 Trading, disclosure, and the market for control

The trading model builds on discrete order flow in the Kyle tradition (Kyle, 1985; Glosten and Milgrom, 1985), in the tractable form of Edmans, Goldstein, et al. (2015), and on the feedback between prices and real decisions (Edmans, Goldstein, et al., 2012; Bond et al., 2012; Goldstein, 2023). Grossman and Hart (1980) make the takeover premium the welfare object for dispersed minority shareholders, and Burkart, Gromb, et al. (1998) show why higher premia protect them.

Against the closest theory papers the position is specific. Kyle and Vila (1991) have an endogenous pooled/revealed split in mixing equilibria but disclaim disclosure requirements on their first page, and their traders disclose nothing at a legal deadline; the defensible contrast is that their split is endogenous, un-keyed, and available only where the trader happens to mix, while the rule-keyed partition here is imposed by a stake trigger and a deadline in every parameter configuration. Collin-Dufresne and Fos (2015) show measured price impact falls when informed blockholders trade against liquidity, and Collin-Dufresne and Fos (2016) fix a trading horizon, but the presence of the informed trader is common knowledge in their model; they name the inference problem and set it aside. Back et al. (2018) embed an engaged blockholder in a continuous-time Kyle model with a fixed horizon at which the stake is disclosed; for them a shorter horizon is a re-parameterisation of noise-trading volatility ("reducing the trading horizon T is isomorphic to reducing noise trading volatility"), there is no acquirer and no premium, and endogenizing the horizon is their own declared out-of-scope extension. Cetemen et al. (2026) study multiple blockholders timing trades and coordinating through market signals inside a pre-disclosure window they leave unmodelled (their own reading of their model), with liquidity the comparative static and no premium object. Corum (2025) has the closest reduced form of the rule, two scalars measuring how much of the stake is tradable before disclosure bites, but collapses threshold and window into them and has no second player. Corum and Levit (2019) put a takeover probability and a disclosure threshold in one model, but the threshold does three jobs at once (liquidity break, 13D trigger, pill trigger), no comparative static in it is stated, and the flagged state never occurs on path. Ordóñez-Calafí and Bernhardt (2022) study the optimal threshold level (crossing reveals the position, so in equilibrium the engaging blockholder's stake is the smaller of the threshold and its unconstrained optimum, the threshold acting as an upper bound on the position and hence on trading profits), and they assert, without a proposition, the interaction between liquidity and the threshold on managerial discipline; the object here is a control outcome and the interaction is proved. Burkart and Lee (2022) have the premium and explicitly hand over the missing piece: they do not endogenize the toehold in anonymous pre-disclosure markets and leave the interaction of pre- and post-disclosure decisions to future work.

2.4 Structural and empirical work on activism

A mature structural literature estimates activism along complementary margins: Gantchev (2013) the sequential costs of campaigns; Johnson and Swem (2021) a dynamic reputation model in which buying and filing are one move; Albuquerque et al. (2022) the binary 13D-versus-13G choice jointly with announcement returns, with liquidity a control variable and no control outcome; Celentano and Levine (2025) an equilibrium model of activism, takeovers, and managerial discipline in which the 5 percent threshold enters once, as a calibration constant, and the window never appears. None of these models the trading-and-disclosure environment itself. Brav et al. (2008) and Greenwood and Schor (2009) document that the targets of engagement campaigns are unusually likely to be acquired, and that fact makes bidder entry the natural control outcome; Becht et al. (2017) show the returns internationally; Zeng (2026) measures what leaks inside the window.

On the February 2024 acceleration itself, two papers precede this one and neither occupies the cell. Polk et al. (2024) use pre-rule data only, a cross-section of self-selected delays offered as a projection with no control group, no post-rule data, and no standard errors anywhere in the paper. Their treatment variable counts calendar days against a rule they themselves state in business days. Trivedi (2026) runs a difference-in-differences on 333 filings with 13G controls and finds a first stage on compliance timing (the share filed within five business days rises by 34.8 percentage points, standard error 13.0) and nulls on spread and Amihud changes that carry no minimum detectable effect; there is no control outcome in the paper, its sampling frame is never stated, and it is not peer reviewed. The cell this paper's empirics occupies (a rule margin, a liquidity split, and a control outcome, with the timing split as the identity claim) remains open, and the specification in Section 6 is written so that every estimate it will report was specified before any of them existed.

3 The model

3.1 Object and partition

The disclosure rule *is* the market's partition. A stake threshold τ and a filing window T split the histories that reach the control node into two cells: a *flagged* cell, on which the filing has landed before the control decision, and a *pooled* cell, on which it has not. Nothing else in the model performs that split, and the two cells are exclusive and exhaustive by construction. The paper's question is how noise-trading intensity κ moves bidder entry and the expected takeover premium once the rule has drawn that line, and how the answer changes when the line is redrawn.

The control-outcome object is the expected engagement-related premium $\Delta^{\text{act}}(\kappa, \tau, T)$. Beside it sit three price-path objects, all of them produced by the model rather than assumed: the run-up path R_d, the cumulative run-up R, and the filing-day jump J. Lower τ is a tighter threshold margin; lower T is a tighter window margin. The two margins are separate policy coordinates and the results below treat them separately, because the model does not deliver the same answer at both.

Three primitives are inherited rather than built here. Order flow is the discrete, ternary-noise structure of the Kyle tradition (Kyle, 1985), in the tractable discrete form used by Edmans, Goldstein, et al. (2015); prices are competitive in the sense of Glosten and Milgrom (1985), set equal to the expectation of the terminal payoff given the public history; and the takeover premium is the welfare object for dispersed minority shareholders for the free-rider reason of Grossman and Hart (1980). What is new is not any of these pieces but the partition that sits between them, and the sections that follow state it exactly.

3.2 Timing: the two rounds

Trading runs over business days $d = 0, \dots, H$, with H finite. The sequence is: pooled round $\rightarrow$ flag or no flag $\rightarrow$ flagged round if applicable $\rightarrow$ bidder decision.

(1) Nature draws the standalone value v and the blockholder's private signal s. The blockholder picks one complete contingent plan j from a finite ordered menu $\mathcal{J}$.

(2) *Round 1: pooled trading.* The plan's stake path executes over $d = 0, \dots, H$. Market makers see pooled order flow and set $P_d^P = \mathbb{E}[Y \mid \mathcal{H}_d^P]$.

(3) *Disclosure node.* The flag lands if and only if $D = 1$: the plan engages, crosses τ at some date $c < \infty$, and $c + T \leq H$. The filing reveals $F = (B^F, a = 1)$.

(4) *Round 2: flagged trading, then the bidder.* If $D = 1$ the blockholder submits the flagged order Q^F, the market sets the flagged price $P^F = P(F, Q^F)$, and then the bidder decides. If $D = 0$ there is no flagged round and the bidder acts on the pooled history.

Parts of this timing are load-bearing, and each is stated as a hypothesis rather than left in the prose, because named proof steps fail without them.

Assumption 1 (Timing and the legal clock)**.** *(a)* No feedback: no within-window re-optimisation. *The blockholder does not re-optimise inside the filing window. There is no feedback from realised order flow or from realised prices into the executed path: $B_j(s,d)$, $q_{jd}(s)$ and Q_j^F are functions of (j, s, d) and of (j, s, τ, T) alone.*

(b) Flag termination: the flag terminates the pooled round. *Pooled trading stops when the filing lands. The flagged round follows the filing, and the bidder acts after the flagged round.*

(c) Legal-clock discipline. *The crossing date c is the first date the path reaches τ; the filing lands exactly at $c + T$; filings truthfully reveal stake and purpose; and only Voice plans cross in the core model.*

(d) Interior crossing. $0 < \Omega(\kappa, \tau, T) < 1$. *This is required only for positive cell mass, never for the structural partition.*

Assumption 1(a) is what makes the executed path a function of the plan and the signal, and named steps of the conditional-independence argument behind Lemma 3 fail without it. Assumption 1(b) is what makes $Q_j^F(s) = b_j^*(s) - B_j^F(s)$ the blockholder's whole residual position; read the other way, the flagged-round step of Proposition 1 fails.

3.3 Primitives

Definition 1 (Primitives: values, signals, noise and the disclosure policy)**.** *The standalone value is $v \sim N(\mu_v, \sigma_v^2)$ and the blockholder's private signal is $s = v + \varepsilon$ with $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ independent of v, so that $\mathbb{E}[v \mid s] = \mu_v + \beta(s - \mu_v)$ with the Gaussian projection $\beta = \sigma_v^2/(\sigma_v^2 + \sigma_\varepsilon^2) \in (0,1)$. The bidder draws a private synergy shock $\xi \sim N(0, \sigma_\xi^2)$, independent of (v, s); $\bar{S}$ is mean bidder synergy and $K > 0$ the bidder's entry cost. Takeover premia without and with engagement are m_0 and m_1, with premium wedge $\Delta_m = m_1 - m_0 > 0$, and $\Delta_V \geq 0$ is the non-takeover value created by engagement. Noise trading is ternary: $z_d \in \{-\bar{z}, 0, +\bar{z}\}$ with $\bar{z} > 0$, $\mathbb{P}(z_d = 0) = 1 - \kappa$ and $\mathbb{P}(z_d = \pm\bar{z}) = \kappa/2$, where $\kappa \in [0,1]$ is noise-trading intensity. The disclosure rule is the pair (τ, T) with $T \in \{1, \ldots, H\}$; the blockholder's initial and maximum stakes are b_0 and $\bar{b}$ with $0 \leq b_0 \leq \bar{b}$.*

Definition 2 (Plans and the legal clock)**.** *The plan menu $\mathcal{J}$ is finite and ordered from least to most aggressive, with $|\mathcal{J}| = J < \infty$. Plan j carries an engagement flag $a_j \in \{0,1\}$, equal to 1 for Voice and 0 for Exit and Hold; the one-round predecessor's four named actions are Exit, Hold, Quiet Voice and Public Voice. Its cumulative pooled stake at day d is $B_j(s,d) \in [0, \bar{b}]$ with $B_j(s,-1) = b_0$; the terminal target is $b_j^*(s) = B_j(s,H)$. The threshold-crossing date is $c_j(s;\tau) = \inf\{d : B_j(s,d) \geq \tau\}$, equal to $+\infty$ if the path never reaches τ, and the legal filing date is $f_j = c_j + T$. The disclosure indicator is $D_j(s;\tau,T) = \mathbf{1}\{a_j = 1,\ c_j < \infty,\ f_j \leq H\}$, so that $D = 1 \Rightarrow a = 1$. The stake at filing is $B_j^F = B_j(s, f_j)$ and the flagged-round order is $Q_j^F = b_j^*(s) - B_j^F$. A finite ordered coarsening Γ maps the stake increment to the pooled order mark, $q_{jd}(s) = \Gamma\big(B_j(s,d) - B_j(s,d-1)\big)$, and observed pooled order flow is $X_d = q_{jd} + z_d$.*

Definition 3 (Information, prices and the control outcome)**.** *The pooled public history is $\mathcal{H}_d^P = (X_0, \ldots, X_d;$ flag landed by $d)$, which is finite. The filing message is $F = (B^F, a{=}1)$, and the flagged tuple is F augmented by the flagged order, $\mathsf{S}_F = (B^F, Q^F, a{=}1)$. The control-node information set is the* public *information at the control node: $\mathcal{I}_H = \mathcal{H}_H^P$ on the pooled cell $\{D = 0\}$ and $\mathcal{I}_H = \mathsf{S}_F$ on the flagged cell $\{D = 1\}$. The bidder's own ξ is private. Write $\mathcal{C}_F$ and $\mathcal{C}_P$ for the flagged and pooled cells; they are exclusive and exhaustive by construction. The engagement posterior is $\pi(\mathcal{I}) = \mathbb{P}(a = 1 \mid \mathcal{I}) \in [0,1]$, equal to 1 on $\mathcal{C}_F$, and the bidder-entry probability is*

$$p(\mathcal{I}) \;=\; 1 - \Phi\left(\frac{P + K + m_0 + \pi\Delta_m - \bar{S}}{\sigma_\xi}\right) \;\in\; (0,1). \tag{1}$$

With B the entry indicator, the terminal shareholder payoff is

$$Y \;=\; (1 - \mathsf{B})\,(v + a\Delta_V) \;+\; \mathsf{B}\,(P + m_0 + a\Delta_m), \tag{2}$$

and prices satisfy the inner fixed point $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$. Two conventions are part of the model rather than notation. First, $P^P_{-1} := \mathbb{E}[Y]$ is the pre-trading pooled price, which is needed whenever $c = 0$ — as $T = H$ forces on every flagged history. Second, $P_{\mathrm{ND}}(\mathcal{H}^P_{f^-}) := P^P_{f^-}$: the not-yet-disclosed price is the last pre-filing pooled price at the same *realised order flow, whose history already carries "flag not landed by $f-1$", and not a never-disclosed counterfactual. Throughout, c^- and f^- denote the business days immediately before the crossing date and the filing date. The price-path objects are the run-up path $R_d = P^P_d - P^P_{c^-}$, the cumulative run-up $R = P^P_{f^-} - P^P_{c^-}$, and the filing-day jump $J = P^F - P_{\mathrm{ND}}$; R_d, R and J are unsigned, and J is not claimed to be κ-invariant.*

The genuine fixed point sits at the control nodes. At an earlier pooled date $d < H$ the price is a tower expectation of already-solved control-node values, with no self-reference; only the control-node map is a fixed point to be solved.

Definition 4 (Objective: the blockholder's payoff U_j)**.** *The blockholder's objective is the expected terminal value of the position the plan builds, net of what it costs to build and to engage:*

$$U_j(s) \;=\; \mathbb{E}\Big[\, b^*_j(s)\, Y \;-\; \mathcal{C}^{\mathrm{trade}}_j \;-\; a_j\, C_j(s) \;\Big|\; s,\, j \,\Big], \tag{3}$$

where $\mathcal{C}^{\mathrm{trade}}_j$ is the execution outlay — increments valued at the pooled prices P^P_d up to the plan's last pooled date, plus $Q^F_j(s)\, P^F$ when $D_j = 1$ — and $C_j(s) \geq 0$ is the engagement cost. Only two properties of (3) *are ever used:* plan-locality*, that U_j depends on j only through the executed stake path, the prices paid on it, the terminal stake, the engagement flag and the cost; and* integrability*, $\mathbb{E}[\max_j |U_j|] < \infty$ under Assumption 2(b).*

Remark 1 (Timing convention for the engagement cost)**.** *Display* (3) *does not date $C_j(s)$. The engagement cost may be booked either on completing the plan or as sunk once the filing has landed. The two readings give the same round-2 comparison on the round-2 deviation set: under the sunk reading the continuation is constant on that set with no further clause, and under the plan-completion reading the equality of costs across the set is what the continuation-cost hypothesis of Proposition 1 buys. Either way the existence result does not depend on the choice; the statement there is written so as not to commit to a reading.*

Assumption 2 (Primitives)**.** *(a)* Independent primitives. *v, ε, ξ and all z_d are mutually independent, and all variances are strictly positive.*

(b) Finite model, amended boundedness. *The plan menu $\mathcal{J}$, the image of Γ (the order-mark support), the noise support $\{-\bar{z}, 0, +\bar{z}\}$ and the calendar horizon H are finite. Prices and payoffs are locally bounded in (s, ϑ) on the maintained parameter set, and $\mathbb{E}\big[\max_{j \in \mathcal{J}} |U_j|\big] < \infty$ for every k in the cutoff polytope Θ of Definition 6 below.*

(c) The primitive table. *The primitives of Definitions 1–4 satisfy the following, which several results below consume as a block.*

(TR-i) Distributional forms and signs. *The distributions of Definition 1 hold as stated, with $\sigma^2_v, \sigma^2_\varepsilon, \sigma^2_\xi > 0$, $K > 0$, $\bar{z} > 0$, $\Delta_m > 0$, $\Delta_V \geq 0$, and*

$$m_0 \;\geq\; 0, \tag{4}$$

so that $\bar{m}(\mathcal{I}) = m_0 + \pi(\mathcal{I})\Delta_m \geq 0$. Restriction (4) *is what makes the inner pricing fixed point exist, be unique and be continuous; dropping it produces both nonexistence and three-root multiplicity in executed counterexamples. The initial stake satisfies $b_0 < \tau$: a pre-existing crossing is outside the core model.*

(TR-ii) Stake-path regularity. *For Voice plans $\partial_d B_j \geq 0$ and $\partial_s B_j \geq 0$; Hold paths are constant and Exit paths weakly decreasing in d. For every plan and every d, the map $s \mapsto B_j(s, d)$ is Borel. Borel regularity is automatic for Voice (monotone in s) and for Hold (constant) and is a genuine addition for Exit, where the primitives supply monotonicity in d only; without it the pooled prices are not defined, because pooled pricing integrates over every type including Exit types. Stake levels are continuum-valued: the finiteness of Assumption 2(b) covers the plan menu, the image of Γ, the noise support and the calendar, not the stake level.*

(TR-iii) Legal-clock objects. *The definitions of c_j, f_j, B_j^F, Q_j^F and b_j^* in Definition 2 hold as stated, together with $D = 1 \Rightarrow a = 1$; $Q^F \geq 0$ for Voice plans; and, at fixed policies, $T' < T$ implies $B^F(T') \leq B^F(T)$ and $Q^F(T') \geq Q^F(T)$.*

(TR-iv) Pricing and entry. *The terminal payoff is* (2)*, prices obey the convention $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$, entry obeys* (1)*, the control-node information set is the fill given in Definition 3, and the two price conventions $P^P_{-1} = \mathbb{E}[Y]$ and $P_{\mathrm{ND}} = P^P_{f^-}$ hold.*

The boundedness clause is stated as integrability because flat global boundedness is inconsistent with the rest of the model: v is Gaussian and the flagged region is unbounded in s under Assumption 4(b), so Y, and with it prices and U_j, is unbounded. Integrability is all any proof below consumes.

3.4 Equilibrium notion

Definition 5 (Cutoff perfect Bayesian equilibrium)**.** *A* cutoff perfect Bayesian equilibrium *is:*

(i) a weakly ordered cutoff vector $k = (k_1 \leq \cdots \leq k_{J-1})$ mapping the signal s into a plan;

(ii) sequentially optimal pooled and flagged components;

(iii) Bayes-consistent beliefs on path;

(iv) competitive pooled and flagged prices at their fixed points;

(v) the bidder-entry rule (1)*;*

(vi) off-path beliefs as limits of full-support perturbations.

The inequalities in (i) are weak, so collapsed action regions — including a collapsed Hold region — are permitted. Existence is a Brouwer argument on the compact ordered polytope Θ for the outer map $\mathcal{T}(k; \vartheta)$ of Section 3.6, at a fixed point $k = \mathcal{T}(k; \vartheta)$. Uniqueness is not *claimed.*

3.5 Standing assumptions

Assumptions 1 and 2 accompany the timing and the primitives above. The four assumptions below are the model's standing hypotheses about the equilibrium object, about identification on flagged histories, about the chord representation behind the pooled cell's liquidity motion, and about the general-equilibrium region. They are not all maintained at once: each result in Sections 4 and 5 names the clauses it consumes. Three of these hypotheses are measured to fail at the calibration the accompanying implementation runs. Section 5.6 states that record in plain words after the results it bears on, and the online appendix carries the full measurements together with the extended discussion of each assumption.

Assumption 3 (The equilibrium object). *(a)* Ordered plans, single crossing. *At every belief and price system, adjacent-plan payoff differences cross zero at most once in s, and the preferred plan is weakly increasing in s.*

(b) Compact outer self-map. *All best-response cutoffs lie in a common compact ordered polytope Θ; the outer map $\mathcal{T}$ of Section 3.6 is continuous and maps Θ into itself.*

(c) Inner pricing regularity. *Each public-history pricing map has a unique fixed point, continuous in beliefs, cutoffs and parameters.*

Clause (c) is mostly a theorem rather than an assumption. Under the regularity stack (4), existence, uniqueness and continuity of the inner pricing fixed point in its belief summaries are proved, and what remains assumed is continuity of the composed price family in the cutoff vector, read as a measurably selected family. The online appendix develops this and records the composition measured discontinuous at the implemented calibration; no result below consumes the cutoff clause. Clauses (a) and (b) are themselves measured to fail at that calibration, and Section 5.6 states the record. None of this moves any result: each clause is a hypothesis, and Proposition 1 remains proved as a conditional.

Assumption 4 (Identification on flagged histories). *(a)* Filing sufficiency. *On flagged histories, $(B^F, Q^F, a{=}1)$ identifies the informed component of the selected plan; conditional on it, the pooled order-flow residual is pure noise, independent of (v, s, ξ).*

(b) On-path composed target. *At a fixed cutoff policy, the composed terminal target $s \mapsto b^*_{j(s)}(s)$ is strictly increasing on the flagged signal region, and this holds for every cutoff vector $k \in \Theta$. Strictness is required only for flag-capable composed targets: passive plans that never flag need not have strictly increasing b^*_j, and there is no backtracking of b^*_j across admissible Voice-plan switches.*

(c) Joint tuple injectivity. *The full map $(j, s) \mapsto (B^F_j, Q^F_j, a_j)$ is injective on the flagged-pair set $\{(j, s) : D_j = 1\}$, including flagged pairs that are not selected on path.*

Clause (a)'s weak identification wording is not enough for the flagged-cell lemma of Section 5: it permits two pairs (j, s) with different pooled paths, which is that lemma's first failure case. Clause (c) is strictly stronger than clause (b), and it is the form the existence argument consumes. Satisfiability is resolved for clause (b): a satisfying menu exists, the pro-rata single-Voice menu with terminal target strictly increasing on all of $\mathbb{R}$, which also satisfies clause (c). The online appendix carries the full discussion, including the failure boundary and the executed checks.

Assumption 5 (The threshold chord restriction)**.** *The pooled posterior law has the symmetric ternary representation*

$$\mathbb{E}[h] \;=\; A_0(\kappa)\,h(0) \;+\; A_{1/2}(\kappa)\,h(\bar{\pi}/2) \;+\; A_1(\kappa)\,h(\bar{\pi}), \tag{5}$$

with $A_0' = A_1' = A_\kappa'$ *and* $A_{1/2}' = -2A_\kappa'$*, and maintained orientation* $C_h(\bar{\pi}) \le 0$ *with* $|C_h|$ *weakly increasing in* $\bar{\pi}$*. The orientation is the weak one, so the case* $C_h(\bar{\pi}) = 0$ *has to be handled explicitly wherever it is consumed. Two further clauses are part of the restriction:*

(τ-i) The kernel depends on the information set only through the engagement posterior: $h(\mathcal{I}) = h(\pi(\mathcal{I}))$*, so the three numbers* $h(0)$*,* $h(\bar{\pi}/2)$*,* $h(\bar{\pi})$ *are well defined and* κ*-free. This is a restriction, not a reading. In the model* $h = \pi\, p(\hat{v}, \pi)$ *is a function of* two *scalars, because* (1) *makes entry depend on the price as well as on the posterior; the clause says the standalone-value channel and the engagement channel do not co-move inside the pooled cell in a way that moves* h *at a fixed posterior.*

(τ-ii) The support and $\bar{\pi}$ are κ-free; only the weights move. *The three points* $\{0, \bar{\pi}/2, \bar{\pi}\}$ *do not vary with* κ*, and* $\bar{\pi}$ *itself is* κ*-free at fixed* (τ, T)*. Without the second half the conclusion of Lemma 4 is false: the derivative gains a term that is first order in* $\bar{\pi}$ *and the vanishing fails.*

Given a κ-invariant three-point support, the derivative restrictions above are equivalent to κ-invariance of the pooled block's total mass and of its unnormalised engagement moment, both of which the model delivers at fixed policies; the entire remaining content of the assumption is the support condition. Whether the two-round pooled cell of Section 3.2 satisfies it is open, and every result conditional on this assumption inherits that conditionality. The online appendix records that the implemented calibration fails it.

Assumption 6 (The bridge and the general-equilibrium region)**.** *(a)* The chord–sensitivity bridge (A(br)). *For two compared thresholds* $\tau' < \tau$ *at fixed policies and a common* κ*:*

(br-i) Representation at both policies. *The symmetric ternary representation* (5) *holds for the pooled class under* τ and *under* τ'*, with chord endpoints* $\bar{\pi}(\tau)$ *and* $\bar{\pi}(\tau')$ *and weight-derivative coefficients* $A_\kappa'(\tau)$ *and* $A_\kappa'(\tau')$*.*

(br-ii) κ-localisation. *At fixed policies all* κ*-dependence of* M_P *sits in the weights of* (5)*: the three support points* $\{0, \bar{\pi}/2, \bar{\pi}\}$ *and the kernel* h as a function of the posterior *do not move with* κ*. Hence* $\partial_\kappa M_P = \Delta_m A_\kappa' C_h(\bar{\pi})$ *exactly, with no composition-through-*κ *remainder. The trailing "hence" is derivable, not assumed. Read against the literal display* (5) *this clause would restate (br-i); it is written against the reading* $h = \pi\, p(\hat{v}, \pi)$*, and it repairs that ambiguity rather than adding a further independent restriction, naming the same object as clause (τ-i) of Assumption 5.*

(br-iii) Coefficient stability across the threshold margin. $|A_\kappa'(\tau')| \le |A_\kappa'(\tau)|$*. The weakest sufficient form is equality: reclassification changes* which *histories are pooled, not the* κ*-responsiveness of the pooled weights.*

(br-iv) Endpoint linkage. $\bar{\pi}$ *is the chord endpoint of* (5) *— the upper support point of the pooled posterior law — and it is a weakly increasing function of the pooled prior*

engagement share $\bar{\pi}_{\mathrm{pr}} = \mathbb{P}(a = 1 \mid D = 0)$*,* the same function at τ and at τ'. *The identity branch* $\bar{\pi} = \bar{\pi}_{\mathrm{pr}}$ *is excluded as degenerate.*

(br-v) Comparability of the chord functional across thresholds. *The chord functional* $C_h(\cdot)$ *— and the kernel* h *it is built from — are the same functions of the posterior at both compared thresholds. Without it, leg 3 of Lemma 5 compares* $|C_h|$ *across two different functionals and the comparison is meaningless:* $h = \pi p$ *with* p *priced off a cell whose composition the threshold moves, so* τ*-invariance of* h *is real content, not bookkeeping.*

(b) General-equilibrium differentiability and contraction. *On a candidate region* $\mathcal{R}$ *the outer map is twice continuously differentiable,* $L_{\mathcal{R}} < 1$*, and the sign of the equilibrium liquidity derivative is constant on* $\mathcal{R}$*.*

Clause (a) is consumed by leg 3 of Lemma 5 and by Part (B) of Theorem 1, and by nothing else; its clause (br-iii) has the least justification behind it and is the one to attack first. Because (br-i) carries the chord representation at both thresholds, the measured failure recorded in the online appendix bears on clause (a) as well.

3.6 The cutoff polytope and the premium objects

Definition 6 (Equilibrium objects: Θ, $\mathcal{T}$, and the general-equilibrium bounds)**.** *The cutoff vector is* $k = (k_1, \ldots, k_{J-1})$ *with* $k_1 \le \cdots \le k_{J-1}$*; when the menu is the four named actions of a one-round predecessor model it maps to that model's triple of cutoffs. The cutoff polytope* Θ *is nonempty, compact and convex, and* ϑ *is the parameter vector. The* outer cutoff best-response map *is* $\mathcal{T}(k;\vartheta)$*, the cutoff vector that represents the best response to the conjecture* k*; it is always written calligraphically, since upright* T *is the filing window. That* $\mathcal{T}$ *is single-valued and continuous and maps* Θ *into itself is the content of Assumption 3(b) rather than of this definition, and the online appendix records that continuity measured to fail at the implemented calibration. On a region* $\mathcal{R}$ *the contraction bound is* $L_{\mathcal{R}} = \sup_{\mathcal{R}} \|D_k \mathcal{T}\|$*, required to be below* 1 *by Assumption 6(b). The strictness coordinates are* $r_\tau = -\tau$ *and* $r_T = -T$*, so that higher* r *is tighter. The direct fixed-policy attenuation margin is*

$$g_r^{PE} \;=\; -\operatorname{sgn}\!\big(d\Delta^{\mathrm{act}}/d\kappa\big)\;\partial_{\kappa r}\Delta^{\mathrm{act}}, \tag{6}$$

with the sign written inline rather than carried by a symbol. The inversion-free derivative bounds are $\bar{k}_x = |\partial_x \mathcal{T}|/(1 - L_{\mathcal{R}})$ *and* $\bar{k}_{\kappa r}$*, and the general-equilibrium remainder bound is*

$$\mathcal{B}_r^{GE} \;=\; |\Delta_{\kappa k}|\,\bar{k}_r \;+\; \big(|\Delta_{kr}| + |\Delta_{kk}|\,\bar{k}_r\big)\bar{k}_\kappa \;+\; |\Delta_k|\,\bar{k}_{\kappa r}. \tag{7}$$

In (7)*,* Δ_k*,* $\Delta_{\kappa k}$*,* Δ_{kr} *and* Δ_{kk} *are the gradient and the cross-partials of* Δ^{act} *in* (k, κ, r) *along the equilibrium branch, and* $\bar{k}_{\kappa r}$ *is the corresponding second-order inversion-free bound; both are constructed in the online appendix. The* dominance-and-contraction region *is* $\mathcal{R}_r$*, with slack* $\eta_r = g_r^{PE} - \mathcal{B}_r^{GE}$*; the region may be empty.*

Definition 7 (Premium decomposition and comparative-statics objects)**.** *The engagement-premium kernel is* $h(\mathcal{I}) = \pi(\mathcal{I})\,p(\mathcal{I})$*, with* $h \ge 0$ *and* $h(0) = 0$*, and the expected engagement-related premium is*

$$\Delta^{\mathrm{act}} \;=\; \Delta_m\,\mathbb{E}\big[h(\mathcal{I}_H)\big] \;\ge\; 0. \tag{8}$$

The cell averages are $M_F = \Delta_m \mathbb{E}[h \mid D = 1]$ and $M_P = \Delta_m \mathbb{E}[h \mid D = 0]$, each defined when its cell has mass. The unconditional flagged weight is $\Omega = \mathbb{P}(D = 1) \in [0, 1]$, which factors as $\Omega = \mathbb{P}(a = 1)\,\omega_a$ with $\omega_a = \mathbb{P}(D = 1 \mid a = 1)$ the disclosed share of engagements. The quantity $\bar{\pi}$ is the upper support point *of the pooled engagement posterior in the representation* (5)*. The pooled engagement share is the* mean *of that law, not $\bar{\pi}$; under Assumption 5 the share is κ-invariant, being a mean-preserving spread, so it cannot be the quantity whose κ-motion Lemma 4 describes, it is strictly below $\bar{\pi}$ in any non-degenerate case, and it equals $\bar{\pi}/2$ only under the level symmetry $A_0 = A_1$. The liquidity sensitivities are $\mathcal{S} = |\partial_\kappa \Delta^{\mathrm{act}}|$ and $\mathcal{S}_P = |\partial_\kappa M_P|$. The chord is*

$$C_h(\bar{\pi}) \;=\; h(0) \;-\; 2h(\bar{\pi}/2) \;+\; h(\bar{\pi}), \tag{9}$$

maintained non-positive with $|C_h|$ weakly increasing in $\bar{\pi}$, and A'_κ is the common derivative of the weights in (5)*, bounded on* $[0, 1]$*. The weight-effect ratios are W_τ and W_T — for instance $W_T = (1 - \Omega(\tau, 5))/(1 - \Omega(\tau, 10))$, at most* 1 *when Ω rises — and the composition-effect ratios are C_τ and C_T, for instance $C_T = \mathcal{S}_P(\tau, 5)/\mathcal{S}_P(\tau, 10)$, which are unsigned. The margin subscript on a composition ratio is always written, because C is otherwise overloaded by the chord C_h, the engagement cost $C_j(s)$ and the cells $\mathcal{C}_F, \mathcal{C}_P$.*

Reading $\bar{\pi}$ as the mean of the pooled posterior law rather than as its upper support point forces a point mass at $\bar{\pi}$ with $A'_\kappa = 0$ and zero interior motion for every kernel. That is degenerate, and it is excluded throughout.

4 Equilibrium existence

Existence is standing ground here, not the claim. The claim the paper is making is the attenuation result of Section 5; what this section supplies is the conditional on which the whole construction rests. The hypothesis set is long, and every clause of it is load-bearing, so the hypotheses are enumerated rather than gathered into a sentence.

Proposition 1 (Existence of a cutoff perfect Bayesian equilibrium)**.** *Assume:*

- *(P-1) Assumption 2(a) (independent primitives);*
- *(P-2) Assumption 2(b) (finite model, integrable objective);*
- *(P-3) Assumption 3(a) (ordered plans, single crossing);*
- *(P-4) Assumption 1(c) (legal-clock discipline);*
- *(P-5) Assumption 3(b) (compact outer self-map), read as in the reading of that clause given below;*
- *(P-6) Assumption 4(c),* joint tuple injectivity*: the map $(j, s) \mapsto (B^F_j, Q^F_j, a_j)$ is injective on the whole flagged-pair set $\{(j, s) : D_j = 1\}$,* including pairs no cutoff vector selects*. This is strictly stronger than the on-path form, Assumption 4(b), and it is the form the argument consumes where it pins* off-path *flagged beliefs;*[1]

[1] The distinction is not cosmetic. An earlier recorded statement of this result carried the on-path form while the proof consumed the joint form. The joint form is stated here because the proof uses it to pin off-path flagged beliefs; the on-path form alone would not suffice. The form is named every time either appears.

(P-7) *Lemma 1 by statement,* with its own hypotheses travelling*;*

(P-8) *the no-feedback timing of Assumption 1(a), read together with the flag-termination clause of Assumption 1(b);*

(P-9) *the* definitional *round-2 action-set hypothesis: the flagged-round action set* is *the plan-generated set* $\{Q^F_{j'}(s)\}$ *taken over menu elements agreeing with* j *on everything already played. This is not a closure condition; the closure form is jointly unsatisfiable with finiteness, by cardinality;*

(P-10) continuation-cost equivalence on that same set*: menu elements sharing* j*'s pooled path up to* $f_j(s)$ *and carrying* $a_{j'} = a_j$ *have the same engagement cost,* $C_{j'}(s) = C_j(s)$*. This holds trivially on any single-Voice menu, where the set is a singleton, and it is live only on menus with two or more Voice plans sharing a pooled path. What it buys, under the plan-completion reading of the timing convention in Remark 1, is this: on that set the flagged price does not move and the flagged order cancels, so the engagement cost is the only thing that can differ between staying and deviating, and at a flagged pair the cutoff vector does not select there is no date-*0 *optimality to fall back on — so without this clause the deviator takes the class member with the smallest cost and requirement (ii) of Definition 5 fails at that node. Under the sunk reading the continuation is constant on the deviation set with no clause at all and this hypothesis is not consumed; it is listed because the statement does not commit to a reading, and it is what makes the conclusion hold under both;*

(P-11) *the restriction* $m_0 \geq 0$ *of* (4)*;*

(P-12) *the blockholder-objective definition of Definition 4, whose* $-a_j C_j(s)$ *term is the display* (3)*, together with the timing convention for* $C_j(s)$ *in Remark 1 — the engagement cost may be booked on completing the plan or as sunk once the filing has landed, the two give the same round-2 comparison on the round-2 deviation set, and the result does not depend on the choice;*

(P-13) *the primitive table restrictions 2(c) that the argument consumes, in particular: the terminal payoff* (2) *with the price convention* $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$ *and the entry rule* (1)*, both from clause (TR-iv); clause (TR-ii)'s Borel regularity for* every *plan including Exit, needed here* directly *and not by way of Lemma 1, whose conclusion is measurability of* D *and the cell map; clause (TR-iii)'s implication* $D = 1 \Rightarrow a = 1$*, the definitions of* c*,* f*,* B^F*,* Q^F *and* b^**, and* $\partial_s B_j \geq 0$ *for Voice; and clause (TR-i)'s distributional forms with* $\Delta_m > 0$*.*

Then at every $\kappa \in [0, 1]$ *a cutoff perfect Bayesian equilibrium over complete contingent plans exists: there is* $k^\star \in \Theta$ *with* $k^\star = \mathcal{T}(k^\star; \vartheta)$*, prices at their inner fixed points and beliefs Bayes-consistent on path, together with the following.*

(i) *Off-path beliefs are limits of* one *full-support perturbation family over plans, fixed once and used to define the price system at every* $k \in \Theta$ *and not only at* $k^\star$ *— because the deviation payoffs that define* $\mathcal{T}$ *read off-path pooled histories — at every pooled history reachable* with positive probability *under some plan profile.*

(ii) *The boundary values of* κ *are handled by* extension*, not by restriction. At* $\kappa \in \{0, 1\}$ *the noise support degenerates to* $\{0\}$ *and to* $\{\pm \bar{z}\}$ *respectively; a pooled history needing a mark outside*

the surviving support is null under every *profile, so it is off nature's path rather than off the players', carries no requirement under Definition 5 (vi), and is read by no step. No cut to* $\kappa \in [0, 1)$ *is taken.*

(iii) Flagged-tuple beliefs are supplied by Assumption 4(c) at every tuple in the image of the flagged-pair map $(j, s) \mapsto (B_j^F, Q_j^F, a_j)$*, on path and off, since the image includes tuples generated by pairs the cutoff vector does not select. The belief is the point mass at the unique generating pair. That point mass is a* version *of the conditional law at every image tuple — the signal is continuous, so a version is what a conditional law is, and any a.e.-equal version serves requirements (iii) and (vi) of Definition 5 equally — and it is the version this equilibrium selects. No tuple outside that image arises, because the round-2 action-set hypothesis (P-9) leaves no off-menu order to produce one.*

(iv) The bidder follows the entry rule (1)*.*

(v) There is a sequentially optimal flagged component at every *flagged pair* (j, s)*, whether or not the cutoff vector selects it. The flagged price is invariant across the round-2 deviation set, since Assumption 4(c) pins the belief at the same* s *and at* $\pi = 1$*, so the flagged order cancels out of the continuation and hypothesis (P-10) makes what remains constant.*

Two readings belong to the statement rather than to the proof. First, Assumption 3(c) is not assumed here*: its existence and uniqueness content is derived from* $m_0 \geq 0$*, its continuity content* in the belief summaries $(\hat{v}, \pi)$ *from the same scalar reduction, and its measurable-selection content from Assumption 4(c) together with clause (TR-ii)'s Borel regularity. What is* not *derived is the cutoff clause of Assumption 3(c): continuity of the composed pooled price family in the cutoff vector* k*, which runs through the conditioning pair* $(\hat{v}, \pi)$ *rather than through the pricing map. That continuity enters only through the reading of Assumption 3(b) given next, and the online appendix records it measured to fail at the implemented calibration. Second,* Assumption 3(b) is read *as asserting that* $\mathcal{T}$ *— under a named tie-break-and-corner selection, without which a correspondence cannot be called continuous — is a well-defined, single-valued, continuous self-map of* Θ*, with* Θ *nonempty as in Definition 6; and, in the* bracket *half, that there is a fixed compact signal interval —* ***the common bracket*** $[s_{\mathrm{lo}}, s_{\mathrm{hi}}]$ *at the given* ϑ*, common in* $k \in \Theta$ *and in the adjacent plan pair, of which* Θ *is the ordered part* ($s_{\mathrm{lo}} \leq k_1 \leq \cdots \leq k_{J-1} \leq s_{\mathrm{hi}}$) *— such that for every* $k \in \Theta$ *and every adjacent plan pair the best-response indifference signal* ***exists*** *and lies in the bracket, which is what makes the finite cutoff vector represent the best-response plan map on the whole Gaussian signal line and not merely inside the bracket. The clause excludes every menu-and-price-system at which some adjacent indifference signal fails to exist — never-selected middle plans as well as the dominated extreme plans; the stronger form is taken so the cutoff vector carries interior data at every* k*, and the exclusion is a named domain narrowing.*

Finally, at any such equilibrium at which Assumption 1(d) holds*, both cells carry strictly positive probability and both are on path; and for the restatement of Assumption 1(d) as a single signal threshold, add H-ord — Voice stake monotonicity across plans — and the upper-set engagement-flag hypothesis (the engagement flags* a_j *equal one* exactly *on an upper set of the ordered menu: there is* j_a *with* $a_j = 1$ *for* $j \geq j_a$ *and* $a_j = 0$ *for* $j < j_a$*).* Uniqueness is not claimed.

Proof sketch. The proof (online appendix) builds the equilibrium object rather than characterising it. Part A fixes a conjectured cutoff vector and shows the conjecture induces a measurable plan-selection map, under which every date-0 object is a deterministic Borel function of the pair (j, s). Part B solves the inner price system at that conjecture, separating a finite pooled layer from a continuum-indexed flagged layer; the scalar reduction behind (4) gives existence, uniqueness and continuity of the inner root, and the flagged layer is priced by inverting the flagged-pair map, which Assumption 4(c) makes injective. Part C supplies beliefs at every history that carries a requirement, on path and off: Bayes where the reaching weight is positive, one fixed full-support perturbation family's limit elsewhere. Part D discharges sequential optimality of the flagged component at every flagged pair, selected or not; the flagged price is invariant across the round-2 deviation set and hypothesis (P-10) makes the remaining continuation constant. Part E builds the outer best-response map on the common bracket. That the map is weakly ordered and carries the compact ordered polytope Θ into itself is derived there from Assumption 3(a); the bracket itself and the continuity of the map are taken from Assumption 3(b) rather than derived. Brouwer's fixed-point theorem then applies, and the six requirements of Definition 5 are assembled at the fixed point, with the boundary values of κ handled by extension. Part F is the addendum: at an equilibrium at which Assumption 1(d) holds, both cells are on path. The full proof carries a closing scope statement of six disclaimers: uniqueness in no form; the addendum conditional at the equilibrium and nowhere else; boundary extension, which is where the withdrawn claim of a belief at every pooled history at $\kappa = 1$ sits; one fixed convention; flagged-order optimality without uniqueness; and two hypotheses that are sufficient rather than necessary and are not verified menu by menu.

5 Liquidity and control outcomes

This section states what the model delivers about the control outcome. The order is the order the model builds it in: the partition and its accounting identities, then what liquidity does inside each cell, then the attenuation theorem, then what survives the move from fixed policies to equilibrium. Section 5.6 then reads the whole stack against the implemented calibration and says plainly what it makes of the antecedents. Each statement carries its conditionality with it; the full numerical records sit in the online appendix.

5.1 The partition and its accounting identities

Lemma 1 (The disclosure partition and the price-path decomposition)**.** *Assume Assumptions 2(a), 2(b), 1(c) and 3(c); the primitive table restrictions of Assumption 2(c); the cutoff selection map of Definition 5 (i); and the two price conventions of Definition 3, namely $P^P_{-1} = \mathbb{E}[Y]$ and $P_{\text{ND}} = P^P_{f^-}$. Assume in addition the two clauses now carried as primitives: Borel regularity of $s \mapsto B_j(s, d)$ for* every *plan including Exit, clause (TR-ii), which is needed only for part (c) below because pooled pricing integrates over every type; and a content for the control-node information set $\mathcal{I}_H$, supplied by Definition 3. Then:*

(a) the disclosure indicator $D = \mathbf{1}\{a = 1,\ c(\tau) + T \leq H\}$ is measurable, and it maps every control-node history into exactly one cell;

(b) for every Voice plan, $f_j \leq H$ if and only if $B_j(s, H - T) \geq \tau$;

(c) each flagged history yields B^F, R_d, R and J, and these satisfy the exact identity

$$P^F - P^P_{c^-} \;=\; R + J. \tag{10}$$

Part (b) is the *clock equivalence* that Lemma 5 and Theorem 1 both consume: it converts a statement about the filing date into a statement about the stake path T days before the horizon, which is what makes window comparisons tractable. Identity (10) is the model counterpart of the timing split the empirics of Section 6 measures: the total price move from just before the crossing to the flagged price decomposes exactly into the cumulative run-up plus the filing-day jump.

Lemma 2 (Two-cell decomposition of the engagement premium)**.** *Assume Lemma 1; the definitions of Definitions 3 and 7; Assumption 3(c), which pins* the *version of* $\mathbb{E}[Y \mid \mathcal{I}]$*; Assumption 2(b) together with clause (TR-i), under which* Δ_m *is finite; and Assumption 2(a), which puts every object on one probability space. Then whenever* $0 < \Omega < 1$,

$$\Delta^{\text{act}} \;=\; \Omega\, M_F \;+\; (1-\Omega)\, M_P. \tag{11}$$

At $\Omega = 1$ *the identity degenerates to* $\Delta^{\text{act}} = M_F$*, and at* $\Omega = 0$ *to* $\Delta^{\text{act}} = M_P$*; in each degenerate case the average over the null cell is* undefined rather than imputed. *That last clause is a non-identification statement, and it is proved rather than asserted.*

5.2 Liquidity in the flagged cell

Lemma 3 (κ-invariance of the flagged cell)**.** *At fixed cutoff* and *execution policies, assume Assumption 2(a); Assumption 2(b)* together with *the primitive table restrictions of Assumption 2(c); Assumption 1(c); Assumption 3(c); Assumption 4(b) in its on-path injective form, consumed almost surely on the flagged set;*[2] *Lemma 1; the no-feedback timing of Assumption 1(a); and* $\Omega > 0$*. Assume also an explicit bidder-entry rule — either* (1) *or any rule with the two properties named in the proof — which is carried as bookkeeping. Then the flagged tuple* $\mathsf{S}_F = (B^F, Q^F, a{=}1)$ *makes the pre-filing pooled history conditionally independent of* (v, s, ξ) *on the flagged set, and consequently the flagged posterior, the flagged price, the entry probability and* M_F *are all invariant to* κ*.*

Assumption 4(a) in its weak identification wording is not sufficient here: it permits two pairs (j, s) with different pooled paths, which is this lemma's first failure case. The form consumed is Assumption 4(b), named as such. This lemma is the theoretical statement behind the empirics' identification claim: the flagged half of the timing split should not move with liquidity, and the pooled half should carry the entire liquidity derivative.

5.3 Liquidity in the pooled cell

Lemma 4 (Interior motion of the pooled premium under the chord restriction)**.** *Assume* A(τ) *(Assumption 5), including both of its clauses: (τ-i), kernel-through-posterior, and (τ-ii), κ-free support* and *κ-free* $\bar{\pi}$*. Assume in addition:* $h(0) = 0$*; κ-free pooled mass and κ-free pooled engagement moment at fixed policies; Lemma 1 by statement; and regularity stated minimally — h continuous*

[2] "Almost surely" is the permissive reading, and it is the only coherent one when B^F is continuum-valued, since then no individual flagged tuple has positive probability.

on $[0, \bar{\pi}]$ and twice differentiable on the open interval $(0, \bar{\pi})$, Darboux's theorem doing the rest, with no continuity of h'' required. For the small-$\bar{\pi}$ corollary (part (c)) assume also a second-order Peano expansion of h at $0+$ and one and the same kernel along the shrinking family; and for the seam at which Lemma 5 consumes this lemma, assume $|A'_\kappa|$ bounded uniformly in $\bar{\pi}$ *along the limit. Then:*

(a) $\partial_\kappa \mathbb{E}_\kappa[h] = A'_\kappa\, C_h(\bar{\pi})$*, exactly;*

(b) $C_h(\bar{\pi}) = \frac{1}{4}\bar{\pi}^2 h''(\zeta)$ *for some* $\zeta \in (0, \bar{\pi})$ *— an identity, not an approximation;*

(c) $C_h = \frac{1}{4}h''(0)\,\bar{\pi}^2 + o(\bar{\pi}^2)$*, so the interior motion vanishes at rate* $\bar{\pi}^2$ *as* $\bar{\pi} \downarrow 0$.

The statement is an "if" and never an "iff": $A'_\kappa = 0$ also kills the interior motion.

The conditionality is the substance of the result, not a caveat on it. Whether the two-round pooled cell of Section 3.2 satisfies the support condition of $\mathrm{A}(\tau)$ is **open**: the online appendix shows that the support condition is the assumption's entire remaining bite, and exhibits a one-round market that satisfies it and a no-disclosure structure that does not.

Lemma 5 (Threshold tightening: nesting, composition, and the pooled sensitivity)**.** *At fixed policies, for $b_0 < \tau' < \tau$ at a common window T and a common κ, with $\Omega(\tau', T) < 1$:*

(leg 1) (unconditional) $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau', T)$*, with every newly flagged history generated by a Voice plan; hence* $\Omega(\tau', T) \geq \Omega(\tau, T)$.

(leg 2) (unconditional) *The pooled engagement* share *falls,* $\bar{\pi}_{\mathrm{pr}}(\tau') \leq \bar{\pi}_{\mathrm{pr}}(\tau)$*, with an exact identity for the difference between the two shares.*

(leg 3) (under A(br)) $\mathcal{S}_P(\tau', T) \leq \mathcal{S}_P(\tau, T)$*, with equality whenever* $C_h(\bar{\pi}(\tau)) = 0$.

Legs 1 and 2 need only: the clock equivalence of Lemma 1(b); the no-feedback timing of Assumption 1(a); fixed policies; $b_0 < \tau' < \tau$ imposed at both *thresholds; Assumptions 2(a) and 1(c); clause (TR-iii)'s implication $D = 1 \Rightarrow a = 1$; and $\Omega(\tau') < 1$. Nestedness is a* conclusion *of leg 1, not a hypothesis of it. Leg 3 additionally needs: Lemma 4 by statement; the maintained* magnitude *monotonicity of $|C_h|$ in $\bar{\pi}$ from Assumption 5, the sign half $C_h \leq 0$ being unused at this leg; and clauses (br-i)–(br-v) of Assumption 6(a).*

Leg 3 inherits the conditionality of Assumption 6(a), and through clause (br-i) it inherits the representation (5) at *both* compared thresholds; the numerical record in the online appendix therefore bears on leg 3 as directly as it bears on Lemma 4. At the implemented calibration leg 3's antecedent is not satisfied, and at that calibration leg 3 says nothing about the implemented pooled cell. Legs 1 and 2 are unaffected: they consume no chord restriction. Clause (br-iii) is the clause with the least justification behind it and is the one to attack first.

5.4 Disclosure attenuation

Theorem 1 (Disclosure attenuation at fixed policies)**.** *At fixed plan and cutoff policies, with $0 < \Omega < 1$ and $\mathcal{S}_P > 0$, and under the hypotheses (T-1)–(T-15) listed below:*

(A) Factorisation. $\mathcal{S} = (1 - \Omega)\,\mathcal{S}_P$*, exactly; and the same factorisation holds for the total-variation aggregate of* Δ^{act} *over any* κ*-grid, with no differentiability required.*

(B) Threshold margin. *Threshold tightening attenuates:*

$$\frac{\mathcal{S}(\tau')}{\mathcal{S}(\tau)} \;=\; W_\tau\, C_\tau \;\le\; 1, \tag{12}$$

because both *ratios lie in* $[0,1]$. *No dominance condition is needed.*

(C) Window margin. *Window tightening attenuates* if and only if $W_T C_T \le 1$, *where* $W_T \le 1$ *is proved — from the clock equivalence of Lemma 1 (b) and the monotone Voice stake path — and* C_T *is* unsigned. *The equivalent form*

$$\frac{\partial_{r_T}\mathcal{S}_P}{\mathcal{S}_P} \;\le\; \frac{\partial_{r_T}\Omega}{1-\Omega} \tag{13}$$

holds on average along the tightening path, *integrated over* $[-T,-T']$, *and exactly in the infinitesimal limit; read pointwise,* (13) *is false.*

The hypotheses are: (T-1) fixed policies; (T-2) Assumption 1(d) at each compared policy; (T-3) $\mathcal{S}_P > 0$*; (T-4) Lemma 2; (T-5) Lemma 3, with its own hypotheses travelling; (T-6) Lemma 1; (T-7) PE-*Ω*, that is* $\partial_\kappa \Omega = 0$ *at fixed policies — derivable rather than assumed, and it is exactly what fails in general equilibrium, which is the term Proposition 2 bounds; (T-8)* κ*-differentiability of* M_P*, which no standing hypothesis of Section 3.5 supplies and which is therefore carried in the proof; (T-9) Assumption 5 at* both *compared policies, with* $\bar{\pi}$ *read as the upper support point; (T-10) Lemma 4; (T-11) clauses (br-i)–(br-v) of Assumption 6(a) at the threshold pair; (T-12) Lemma 5; (T-13) the no-feedback timing of Assumption 1(a); (T-14) a smooth window interpolation, for the local form* (13) *only; and (T-15) threshold-side smoothness, which has been confirmed non-load-bearing.* No unconditional window sign is claimed.

Part (A) is the weight effect in its exact form: the liquidity sensitivity of the expected engagement-related premium is the pooled cell's sensitivity, scaled by the pooled cell's weight. Part (B) is the threshold-margin result and needs no dominance condition, because the weight ratio and the composition ratio both lie in $[0,1]$. Part (C) is the window-margin result, and it is an "if and only if" rather than a sign: the composition ratio C_T is unsigned, so the model does not deliver window-margin attenuation as a theorem. The two margins of the disclosure rule therefore enter the policy discussion with different epistemic status, and the February 2024 acceleration, a window change, lands on the margin where the theory signs nothing.

The disclosure-regime comparison, and why it is not a window test. A separate numerical comparison is sometimes read as evidence about the window, and it is not one. In the one-round predecessor model the expected engagement-related premium can be computed twice: once with the market seeing the disclosure flag alongside order flow, once with order flow alone. The cutoffs k_1 and k_0 stay at the baseline equilibrium throughout and only the disclosure cutoff k_D moves; moving it across four values generates the flagged weights $\Omega = 0.037$, 0.129, 0.286 and 0.50. The ratio of liquidity sensitivities (flagged divided by pooled, total variation over $\kappa \in [0.15, 0.85]$) is 1.064, 1.184 and 1.136 at the first three of those weights, falling to 0.378 at $\Omega = 0.50$, with the sign crossing located by bisection at $\Omega^\star = 0.343$. Above one, the flag makes premia *more* liquidity-sensitive. Figure 1 plots the two curves at the baseline. This is a *disclosure-regime* comparison (the flag observed versus hidden, at a fixed filing window), and its ratios are regime-comparison composition outcomes: they are not $W_T C_T$, they measure no window pair, and the predecessor model has no window to vary. The comparison earns its place for one reason: it shows that a composition factor can exceed one, which is what motivates the genuine window-margin "if and only if" of Part (C) and warns against reading attenuation into any flag comparison.

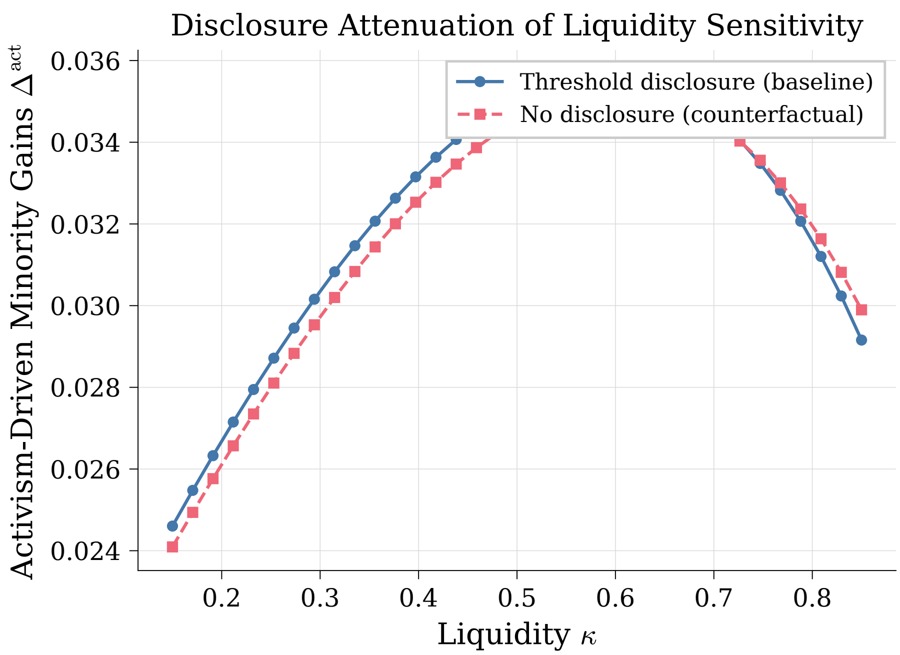

Figure 1: The disclosure-regime comparison in the one-round predecessor model, at fixed cutoffs ($\Omega = 0.037$). The solid line with circle markers prices the blockholder's engagement-driven premium with the market seeing order flow and the disclosure flag (threshold disclosure, baseline); the dashed line with square markers prices it from order flow alone (no disclosure, counterfactual). The two curves cross near $\kappa = 0.73$. At this baseline the flagged cell is *more* liquidity-sensitive than the pooled one (total variation ratio 1.064), so this comparison refutes attenuation in κ for the regime margin at this weight. It is a composition fact, not a window test; the window record is Table 1.

5.5 From fixed policies to equilibrium

Theorem 1 holds at fixed policies, and hypothesis (T-7) is exactly what fails once the cutoff vector is allowed to move with κ. The next result asks what survives that move. It carries the region as a *named hypothesis* rather than exhibiting one.

Proposition 2 (The dominance-and-contraction implication in general equilibrium)**.** *Assume:*

(C-1) the along-the-path contraction of Assumption 6(b), $L_{\mathcal{R}} < 1$, in one fixed norm convention *— an induced operator norm with its dual pairings; a mismatched pairing silently voids the implication;*

(C-2) $\mathcal{R}_r$ relatively open in both *coordinates, with $\kappa \notin \{0, 1\}$;*

(C-3) an interior single-branch equilibrium;

(C-4) twice continuous differentiability of Δ^{act} *in* (k, κ, r)*;*

(C-5) a non-vanishing equilibrium *liquidity derivative — the equilibrium sensitivity* $\mathcal{S}^{GE}$*, which is distinguished from the fixed-policy sensitivity* $\mathcal{S}$ *of Definition 7;*

(C-6) strict dominance $g_r^{PE} > \mathcal{B}_r^{GE}$ *on the region, with* g_r^{PE} *as in* (6) *and* $\mathcal{B}_r^{GE}$ *as in* (7)*;*

(C-7) the threshold *margin* r_τ *only: the window coordinate is an integer, and nothing local is claimed there.*

Then the fixed-policy attenuation sign survives in equilibrium on the region:

$$\partial_r \mathcal{S}^{GE} \;\leq\; -\eta_r \;<\; 0. \tag{14}$$

The sign-coherence hypothesis — that the fixed-policy attenuation sign agrees with the equilibrium sign — is confirmed unused in (14)*: its work is one of naming, fixing when that conclusion may be called a survival of the fixed-policy sign. The sign-constancy clause of Assumption 6(b) is likewise unused.*

5.6 The model at the implemented calibration

The results above are conditionals, and this subsection says plainly what the implemented calibration makes of their antecedents. The online appendix carries the full measurements.

Existence. The two main hypotheses about the equilibrium object are measured to fail at the calibration the implementation runs. The single-crossing clause fails at two independently found loci: at one, an adjacent-plan payoff difference crosses zero three times on an open set of the second cutoff; at the other, the preferred plan decreases in the signal across a cell edge. The outer map's continuity fails for the polytope the proof constructs, with measured jumps of 6.33×10^{-3} to 2.83×10^{-2} across steps in the cutoff of at most 2×10^{-9} at the baseline node, and jumps reaching 0.16 at a second node. In plain words: the proof needs the outer best-response map to be continuous and single-valued, and at the implemented calibration that map has measured jumps and stretches where no weakly increasing selection exists at all. Nonexistence is neither claimed nor shown: twenty-three of twenty-seven equilibrium-sweep nodes converge, the remaining four stay unresolved after thirty seeds each, and a discontinuous self-map may still have fixed points.

The chord restriction. The restriction behind the pooled-cell results fails at the same calibration. The pooled cell's engagement-posterior law, enumerated exactly over all 4^{H+1} order-flow paths at $H = 10$, at two hundred nodes, carries between 23 and 767 distinct posterior values where the restriction's representation asks for three, at every one of the 180 non-degenerate nodes; the remaining twenty nodes are degenerate and decide nothing. The interior motion of the pooled premium, leg 3 of the threshold-nesting result and Part (B) of the attenuation theorem remain proved as conditionals; at this calibration they say nothing about the implemented pooled cell. The open question concerns the restriction's domain: a different menu, a different horizon, or a different calibration could still satisfy the support condition.

The window record. At frozen policies, with cutoffs pinned at the baseline equilibrium, the flagged weight is $\Omega = 13.84$ percent, the disclosed share of engagements is $\omega_a = 61.15$ percent, and the cell averages are $M_F = 0.553$ and $M_P = 0.223$ premium percentage points. Cutting the window from $T = 10$ to $T = 5$ then cuts the premium's liquidity sensitivity to between 18 and 77 percent of its long-window value at every checked node, and the composition leg carries almost all of it: C_T runs from 0.21 to 0.79 while the weight leg shaves 3 to 14 percent (Table 1). The same record confirms the flagged cell's invariance exactly, with every flagged object invariant in κ to machine zero over the grid while the pooled M_P moves 0.0722 premium percentage points, and gives the threshold margin the same direction, with zero of eight compared pairs above one. Two caveats travel with the record. At $H = 10$ the long window is the corner $T = H$; the $H = 12$ column of the table, where $T = 10$ is strictly interior, points the same way but travels the chord closed form whose support condition fails at this calibration, so it is directional corroboration rather than a second independent magnitude. And the record is node evidence at one calibration, not a sign theorem: Part (C) claims none. At this calibration the model's directional prediction for the February 2024 acceleration is attenuation, and the empirics of Section 6 treat that as one of two live branches, not as a derived sign.

Table 1: The window-margin record at fixed policies: cutting the window from $T = 10$ to $T = 5$ at the implemented calibration. Weight leg W_T, composition leg C_T, product $W_T C_T \leq 1$ (attenuation), with no τ node above one. Verified on the numerical grid; sensitivity measured as the total variation of Δ^{act} over the κ grid. The $H = 12$ column re-runs the comparison with $T = 10$ strictly interior to the horizon, where the 0.9-quantile node returns 1.0000 because the τ ladder stops biting and the two flagged weights coincide, not because the product crosses one.

τ quantile	W_T	C_T	$W_T C_T$	$W_T C_T$ at $H = 12$
0.1	0.8559	0.2124	**0.1818**	0.1099
0.3	0.8559	0.2124	**0.1818**	0.1099
0.5	0.8622	0.2384	**0.2055**	0.2406
0.7	0.9176	0.4686	**0.4299**	0.5772
0.9	0.9730	0.7939	**0.7724**	1.0000

The equilibrium region. For the move from fixed policies to equilibrium, eighteen of eighty grid nodes satisfy the two pointwise inequalities, the contraction bound below one and positive dominance slack, with the largest contiguous block, nine nodes, at $T = 5$. A node is a pointwise numerical fact, not evidence of a region: whether any parameter vector satisfies the full antecedent of Proposition 2 is open.

6 Empirics: motivating facts and the written specification

The empirical half of the paper has two layers with different statuses, and the draft keeps them apart. The motivating facts in Section 6.1 have been run: they are descriptives computed from the seeded filing sample, with the sample stated exactly and no standard errors claimed. The specification in Section 6.2 is prospective: it states the variables, windows, samples, standard errors, placebos, and decision rules before any estimate exists. The timing split is the specification's default headline; final selection remains pending in Section 6.3.

6.1 Motivating facts: the delay distribution before and after the acceleration

The anchor is the February 2024 acceleration: the initial Schedule 13D deadline moved from ten calendar days to five business days for trigger dates from 2024-02-05 (U.S. Securities and Exchange Commission, 2023). The first question is whether filers moved at all. They did. Drawing a seeded sample of roughly one hundred fifty initial 13D filings from each of two windows (2023Q2–Q3, the last full pre-rule quarters, and 2024Q3–Q4, well after the effective date) and measuring the trigger-to-filing delay in business days with the federal-holiday calendar, the distribution compresses sharply (Table 2 and Figure 2).

Table 2: The trigger-to-filing delay in business days, before and after the February 2024 acceleration. Seeded samples of initial 13D filings drawn from EDGAR, with the trigger date read from the filing itself and the delay counted in business days on the federal-holiday calendar. Of the 300 sampled filings (150 in each window), 198 carry a trigger date that parses: 102 pre and 96 post, which are the parse rates of 0.68 and 0.64 against the 150 drawn per window. An admissible band of 0 to 60 business days then screens out ten of those, 4 pre (delays of 129, 143, 526 and 567 business days) and 6 post (delays of 62, 73, 133, 262, 407 and 1,365 business days), and the statistics below are computed on the 188 that remain (98 pre, 90 post). Because the parse rates are below one, every count is a floor. Descriptive statistics; no standard errors are claimed.

Window	n	Mean	Median	p90	Share $\leq$ 5 bd	Share $\leq$ 10 bd
Pre (2023Q2–Q3, 10-calendar-day rule)	98	9.63	7.0	23.0	35.7%	80.6%
Post (2024Q3–Q4, 5-business-day rule)	90	6.40	5.0	11.1	75.6%	88.9%

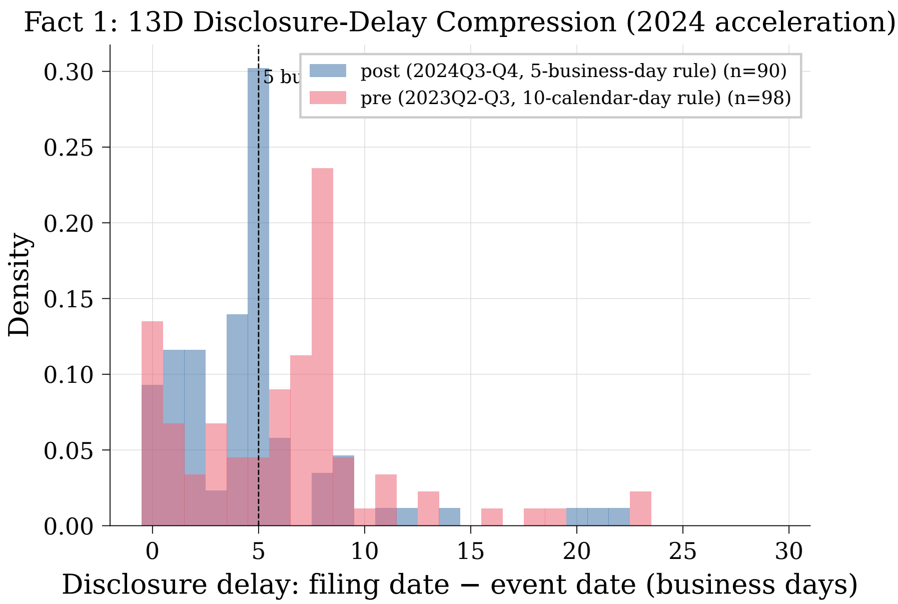

Figure 2: The filing-delay distribution, pre (2023Q2–Q3) versus post (2024Q3–Q4) acceleration, in business days. The mass under the old ten-day regime piles between five and ten business days, with a long right tail of late filers; under the five-business-day rule the mass concentrates inside the new deadline. Descriptives from the seeded samples of Table 2.

Three readings of the table, each bounded. First, the compression is real and large: the median delay falls from seven business days to five, and the share of filings inside the new five-day deadline more than doubles, from 35.7 to 75.6 percent. Second, the pre-rule distribution was not bimodal. It was already spread across the old window in the way Bebchuk et al. (2013) document on filings by engaging investors: the mass rises into the deadline bucket, where 45.3 percent of their filings sit at eight to ten business days, with 16.5 percent inside three business days and a tail of 7.1 percent beyond fifteen. The acceleration bit a population that was heterogeneous in exactly the dimension the rule targets; the design below exploits that heterogeneity as a dose. Third, the numbers agree in kind with the Commission's own arithmetic: the release expected earlier filing for about 59 percent of timely 13D reports and put about 29 percent of 2022's initial filings already inside the amended deadline, against our 35.7 percent already inside before the rule. Different samples and different windows, but the same direction (U.S. Securities and Exchange Commission, 2023). The honest caveats travel with the table: the samples are seeded draws, not the universe; the parse rates are floors; ten filings whose measured delay falls outside a nought-to-sixty business-day band are screened out and counted in the table note rather than cut silently; and these are descriptives, with no standard errors, because none are claimed. The delay is counted in business days throughout; counting calendar days against a business-day rule, as Polk et al. (2024) do, misclassifies the 48 percent of their sample that piles at their day-ten marker.

6.2 The written specification for the full empirics

This subsection summarises a design fixed before estimation; nothing in it has been run. Every estimate the December draft will report is specified here or in the underlying document, and the rule the design imposes on itself is that no specification may change after an estimate is seen, and that a null is reported with its minimum detectable effect rather than as "no effect".

Sample and split. The universe is all initial Schedule 13D filings from 2022-01-01 to 2025-12-31, enumerated from EDGAR quarterly indexes, matched to CRSP. The treatment split is on the *trigger* date, not the filing date: the deadline attaches to the crossing, so a block crossed on 2024-01-30 and filed on 2024-02-08 belonged to the old regime. Filings that straddle are excluded and counted. The main pre window is 2022-01-01 through 2023-10-09. The adoption date, 2023-10-10, begins a separately reported adoption-to-effect stub through 2024-02-04; the stub is not pooled into "pre". The structured-data mandate on 2024-12-18 is a second measurement break, so the main sample ends on 2024-12-17 and 2025 is reported as an extension with its own parse-rate table. The EDGAR dissemination cut-off moved from 5:30 pm to 10:00 pm ET on the same date the window changed, which shifts the flagged day by one for part of the post sample; the effective filing date FD^* is constructed to absorb this, and re-running with $\mathrm{FD}^* \equiv \mathrm{FD}$ is a required robustness row. A full re-fetch and re-parse with the current parser is a stated precondition for any estimate, and every count in the design is a floor until it has run.

(a) The timing split. For every filing, two abnormal returns from a market model estimated over $[\mathrm{TD} - 252, \mathrm{TD} - 22]$: the *run-up*, from the trigger to the day before the flag lands, and the *jump*, on the filing day (a three-day window around the effective filing date). This is Lemma 1's identity (10) and Lemma 3's invariance turned into a regression pair. The partition test (H1) is sign-free: the run-up's liquidity slope should be non-zero, the jump's zero, and the two different. It is estimated as one stacked regression with a flagged-half interaction, so the difference carries a standard error; inference is two-way clustered on subject firm and on calendar month, with a wild cluster bootstrap on the month dimension and the more conservative of the two reported. The reform test (H2) asks whether the run-up's liquidity slope changed after February 2024, on a fixed five-trading-day window from the trigger so the outcome's length does not change mechanically with the rule; both signs are written out in advance: attenuation (the shorter window moves mass to the κ-invariant flagged cell and flattens the composite slope) and amplification (the same informed trading concentrated into fewer days steepens it). One check can kill the identity: if the jump's slope moves the same way as the run-up's, there is no partition to write about. Liquidity is a pre-treatment Amihud-type illiquidity measure, inverted and standardised within quarter, with the orientation printed in every table note. The run-up *path* from trigger to filing, cut by liquidity tercile and period, is reported as a figure in the house style, descriptive only; the benchmark shape is a 0.9 percent abnormal return on the trigger date and 1.6 percent on the day after (Polk et al., 2024), with the run-up beginning on the trigger date itself (Zeng, 2026).

(b) The bindingness dose. The rule binds filers unevenly: those already filing inside five business days cannot be moved by it. The primary dose is a filer attribute built from pre-period behaviour only and is computed for filers with at least two pre-period initial 13Ds: the share whose delay exceeded five business days, in $[0, 1]$. For a single-filing filer, a leave-one-out pre-period stratum mean, by filer type, size tercile and illiquidity tercile, is imputed only in a robustness row and never in the primary estimate. The dose specification interacts the dose with Post and clusters on filer as well. Its first stage is mechanical (a fast filer cannot get faster) and is labelled a validity check, not a finding; the only working-paper estimate of the same first stage is Trivedi (2026)'s compliance share, +0.348 with standard error 0.130, whose own primary outcome, mean lag, is null, and whose sampling frame is never stated. Because slow filers are plausibly smaller and less liquid, the correlation between dose and liquidity is reported, and the dose specification is re-run within liquidity tercile.

(c) Stake at filing. The percent of the class printed on the 13D is the empirical counterpart of the model's stake at filing B^F, an object the two-round model produces and the predecessor model could not. The specification asks whether the stake at filing fell after the acceleration and whether the fall is ordered by liquidity. One hazard is stated before the estimate because it points the other way: in the cross-section of delays, Bebchuk et al. (2013) find the *negative* gradient: filers who take longer end up with smaller stakes, 0.6 percentage points per ten days. The cross-section of delays is therefore never read as a window effect; only the reform is. The historical days–stake gradient implies a mechanical stake change of roughly 0.12 percentage points from the two-day median-delay cut, against the stake leg's minimum detectable effect of 0.85 percentage points on the post-re-parse projection, so a detectable estimate would have to be behavioural rather than arithmetic, and the paper would have to say which.

(d) The bounded null. The Commission's own tables cap what the accumulation channel can do. Classifying 2011–2021 non-corporate-action initial 13Ds by how much of the reported stake was already accumulated by the amended five-business-day deadline: 80 percent of campaigns were unconstrained; 20 percent would have had some accumulation cut; 3 percent would have had at least a tenth of the stake cut; 1 percent at least a quarter (U.S. Securities and Exchange Commission, 2023). If only constrained campaigns can have their bid outcomes moved by the rule, the aggregate effect on the twelve-month bid hazard is capped by the constrained share, and the design reports the whole ladder (20, 3 and 1 percentage points) rather than picking one. The 3-point rung is a judgement about what counts as a materially cut campaign, and it is stated as such. The bound covers the accumulation channel only; it is not the aggregate footprint of the rule, and the dollar figures that travel with it in the release are the Commission's own, and the Commission disclaims them.

(e) The matched difference-in-differences on the bid hazard. The control-outcome leg asks whether a bid arrives within twelve months of the trigger, for 13D targets against *never-13D* firms matched three-to-one on size, illiquidity, industry and quarter, each control inheriting its treated firm's trigger date as a pseudo-trigger so both are measured over the identical calendar window. Never-13D rather than 13G controls: 13G filers disclaim control intent, and selection between the two forms is itself plausibly responsive to the rule. Bid arrival is coded from tender-offer and merger filings, with a thirty-filing hand audit done blind. Base rates anchor the power arithmetic: 18.1 percent of campaign targets are acquired within twelve months against 7.2 percent of matched controls (Greenwood and Schor, 2009). The power arithmetic gives minimum detectable effects of 5.8 percentage points on today's counts, 4.4 percentage points for S1 after the re-parse, and 9.9 percentage points for S2. The latter two use the optimistic projection $N \approx 4{,}950$, $w \approx 0.394$;

they are projections, not estimates. The MDE exceeds the 3-point ceiling of the bounded null, so the difference-in-differences cannot separate the accumulation effect from zero even at that ceiling. Only an estimate's excess above 3 percentage points can be attributed to incumbent defence or selection; the first 3 points remain compatible with accumulation.

(f) Bidder entry by liquidity. Within the 13D sample, the bidder-entry leg estimates the liquidity interaction

$$\mathrm{BID12}_i = \alpha + \delta(\mathrm{LIQ}_i \times \mathrm{Post}_i) + \beta\,\mathrm{LIQ}_i + \gamma\,\mathrm{Post}_i + X_i'\theta + \mathrm{FE} + \varepsilon_i.$$

Against the matched never-13D controls it estimates the matched triple difference

$$\mathrm{BID12}_i = \alpha + \tau(\mathrm{Treat}_i \times \mathrm{Post}_i \times \mathrm{LIQ}_i) + \text{all lower-order interactions} + X_i'\theta + \delta_{\text{match}} + \varepsilon_i.$$

The coefficient τ asks whether the reform's effect on bidder entry differs by liquidity. A rule-of-thumb power calculation gives projected MDEs of about 8.8 percentage points for S1 and 19.8 percentage points for S2. These are weakly powered projections, and the realised variance will replace the rule of thumb before an estimate is interpreted.

(g) The referee checklist. Control group or bounded null: both, as above. Confounds: ten named ones with a stated handling each, including the three the literature specifically warns about: the EDGAR cut-off moving on the same date as the window, calendar-day screens that stop being neutral under a business-day rule, and the incumbent-defence channel, which runs opposite to accumulation. Rights-plan coverage is untested: the proposed moderator uses best-effort EDGAR searches for 8-K Item 3.03 and 8-A12B; if coverage is insufficient, the defence channel is retained with its signed-bias fallback. Power: computed from counts on disk plus stated variance assumptions, with the arithmetic shown; the timing split detects 1.1–2.3 percentage points per standard deviation of illiquidity after the re-parse, against a mean pre-rule run-up of 2.8 percent (Zeng, 2026), and anything smaller is reported as a bounded null. Placebos: 568 holiday-adjusted pseudo-reform dates before the adoption, every one, plus a pseudo-trigger placebo one quarter before the real crossing and a descriptive 13G placebo whose deadlines did not change until September 2024. Pre-trends: quarterly liquidity-slope estimates over the seven pre-adoption quarters, joint F-test, and a fixed blocking rule: a p-value below 0.10 removes causal language from the leg. Parser validation remains incomplete: the available pipeline output is pre-XML and the full-universe re-parse has not run; both limitations are reported.

6.3 Headline shell

The headline empirical result is a choice that has not been made, and this subsection is a shell: it fixes what the headline subsection must contain, lists the candidate headlines with what each would claim and what would kill each, and marks the choice pending rather than picking one.
[**headline choice pending**]

Candidate A: the partition test as headline. Claim: the run-up's liquidity slope is non-zero, the jump's is zero, and the difference is measured with a standard error: the paper's identity in one regression pair, sign-free and needing no theory input. What it needs: the re-parsed sample and the stacked specification of (a). What kills it: a jump slope that moves with liquidity like the run-up's, which would refute the partition itself and is reported first if it happens. Scope: it is an identity claim, not a reform estimate, and a referee will ask what the reform did.

Candidate B: the reform's slope change as headline. Claim: the run-up's liquidity slope changed after February 2024, in the direction the window record of Table 1 points (attenuation at every checked node, composition leg carrying it). What it needs: the same sample, the fixed-length run-up, and a decision on how much weight to put on a theory direction whose antecedent $A(\tau)$ is measured to fail at the implemented calibration, so the design keeps both branches live and pre-specifies what each sign means. What kills it: a bounded null (the minimum detectable effect is 1.1–2.3 percentage points on the post-re-parse projection), which is a reportable outcome rather than a failure. Scope: it is the candidate most directly about the anchor, and the one most exposed to the theory's honest conditionality.

Candidate C: the bounded null as headline. Claim: the accumulation channel's aggregate effect on the bid hazard is capped at about three percentage points by the Commission's own tables, the matched design cannot separately detect even the ceiling, and that arithmetic, not a point estimate, is the paper's answer to what the acceleration could do to control outcomes. What it needs: the matched sample and the ladder of rungs. What kills it: nothing in the data; the risk is presentation, since it reports a ceiling rather than an effect. Scope: it is the most defensible statement and the least newsy one, and it can be combined with either A or B as a supporting leg rather than standing alone.

The choice among A, B, and C is mine, and the December draft's headline subsection will be written into this shell once it is made.

7 Conclusion

The disclosure rule is the market's partition, and this paper takes the identity literally. A stake threshold and a filing window split the histories that reach a control decision into a flagged cell and a pooled one; the window is a primitive of the model rather than a rescaling of noise, and the stake at filing is an object the model produces. On that ground the paper proves what it can and records where it cannot. Existence is a conditional whose two main hypotheses about the equilibrium object are measured to fail at the implemented calibration, stated once, in plain words, because a reader deserves to know exactly what the proposition does and does not cover. The attenuation theorem separates the rule's two margins honestly: the threshold margin attenuates with no dominance condition; the window margin attenuates if and only if a weight ratio times an unsigned composition ratio is at most one. The general-equilibrium survival of the fixed-policy sign is an implication with the region as a named hypothesis, plus numerical nodes. The numerical layer reads the same stack against the implemented calibration and reports an attenuation-direction window record alongside the measured failure of the chord restriction that would close the loop; the two facts sit in the same section because they are both true. The scope is stated as plainly as the results: the paper does not claim a global window-margin attenuation sign, κ-invariance of the filing-day jump, equilibrium uniqueness, a nonempty general-equilibrium region as a theorem, endogenous filing before the deadline, noisy or partially revealing flagged-round trading, continuous-time execution, or welfare or optimal rule design; nor that the predecessor model's non-monotone result for minority gains survives in the two-round model, nor that the predecessor calibration at $\Omega \approx 0.037$ is economically meaningful.

The empirics rest on the February 2024 acceleration. The established descriptive fact is that filers moved: the median delay fell from seven business days to five, and the share inside the new deadline more than doubled. The specification for everything beyond that is written: a timing split that tests the partition directly, a bindingness dose, a stake-at-filing leg, a bounded null from the Commission's own tables, a matched design on the bid hazard whose honest deliverable is already known to be a ceiling, and a bidder-entry-by-liquidity leg. The headline choice is recorded as pending.

Three open questions carry the research forward: whether any two-round pooled cell satisfies the chord restriction's support condition, which is the largest single conditionality in the stack; whether the existence assumptions can be satisfied on a relevant menu, including the incentive-compatibility question for a fully separating plan; and whether the scoped continuity problem in Assumption 3(b) can be repaired constructively. Alongside them stands one open calibration input that is nowhere load-bearing: the disclosed share of engagements ω_a, for which no dataset in the literature offers an anchor, and which the paper does not invent. Each question is about the domain of an assumption this paper states rather than hides, which is where a theory of a legal rule should leave its unfinished work.

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